DevOps Lab Program- 4

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Project Structure:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ tree -I node_modules/
.
├── build
├── Dockerfile
├── package.json
├── package-lock.json
└── src
    └── index.js

3 directories, 4 files
```

<u>Steps for execution:</u>

Step 1:

- Create a folder called program-4
- Run *npm init* to initialize a [Node.js](http://node.js) application
- Run *npm install express* to install express
- Create a Dockerfile using *touch Dockerfile*

Step 2:

Create a folder called src and add a file index.js with the following code:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ cat src/index.js
const express = require('express');
const app = express();
const port = 3000;

app.get("/", (req,res) => {
        res.send("Hello from multi-stage Docker!");
});

app.listen(port, () => {
        console.log(`Running server at port ${port}`);
});
```

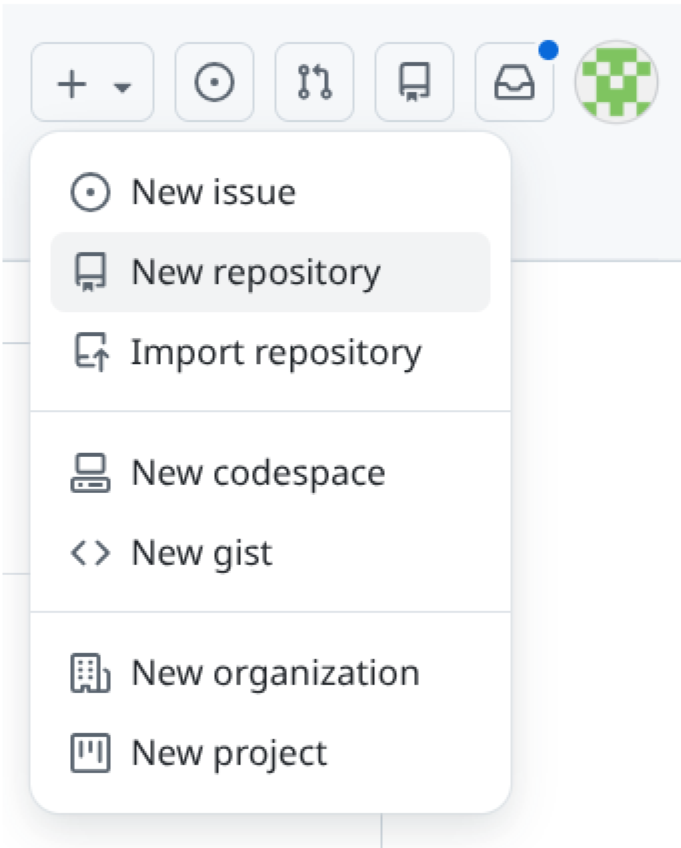
Step 3:
Modify the package. json file:
(Add the following scripts highlighted)

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ cat package.json
{
  "name": "program-2",
  "version": "1.0.0",
  "description": "Program 2 of DevOps lab",
  "main": "index.js",
  "scripts": {
    "start": "node dist/index.js",
    "build": "mkdir -p dist && cp src/* dist/",
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  }
}
```

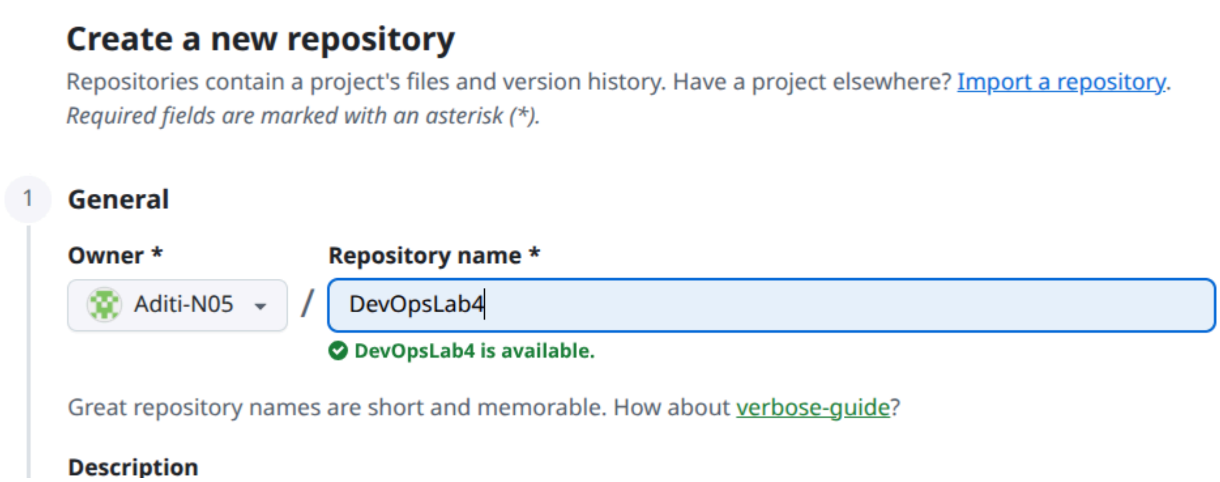
Step 4:
Write the Dockerfile:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ cat Dockerfile
FROM node:20-alpine AS builder

WORKDIR /app

COPY package.json package-lock.json ./
RUN npm install

COPY . .

RUN npm run build

FROM node:20-alpine

WORKDIR /app

COPY --from=builder /app/package.json ./
COPY --from=builder /app/package-lock.json ./
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules

EXPOSE 3000
CMD ["node", "dist/index.js"]
```

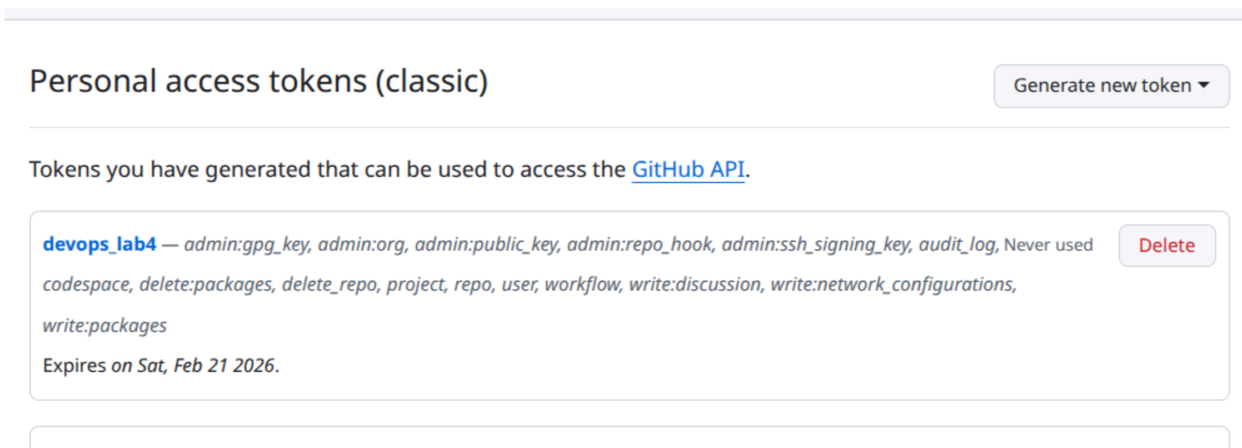
Step 5:
Create a new repository on GitHub:
Login, click on Create New → New Repository



Give the repository a name:



Step 6:
Create a Personal Access Token (PAT):
Go to Settings → Developer Settings → Personal Access Tokens → Tokens (classic) → click on Generate new token → give name and permissions → generate the token → copy the key



Step 7:
Initialize git repository locally:

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1rv24mc005_aditi/dockerFolder/program-4/.git/
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ git add .
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ git commit -m "Initial commit"
```

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ git branch -M main
```

Step 8: Link the local repository to the remote repository:
git remote add origin https://github.com/<username>/<repo-name>.git

Step 9: Push the code to GitHub:

(Enter GitHub username, and the PAT for password when prompted in the terminal)

```
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ git branch -M main
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ git remote add origin https://github.com/Aditi-N05/DevOpsLab4.git
1rv24mc005_aditi@aditi-Inspiron-3584:~/dockerFolder/program-4$ git push origin main
Username for 'https://github.com': Aditi-N05
Password for 'https://Aditi-N05@github.com':
Enumerating objects: 723, done.
Counting objects: 100% (723/723), done.
Delta compression using up to 4 threads
Compressing objects: 100% (667/667), done.
Writing objects: 100% (723/723), 732.37 KiB | 1.98 MiB/s, done.
Total 723 (delta 144), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (144/144), done.
To https://github.com/Aditi-N05/DevOpsLab4.git
 * [new branch]      main -> main
```

Step 10: Check the repository on GitHub to verify that the code has been pushed: